MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

AMPLIFIER, POWER, ANALOG, KEY CODE SES

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic power amplifier module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.42 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be SES.

3.5.2 Size. The module size shall be 1C.

3.5.3 Weight. The module shall have a maximum weight of 115 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall not extend below the interface plane on the B and C increments.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/184-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for power supply voltages of ±15.0 ±0.75 Vdc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

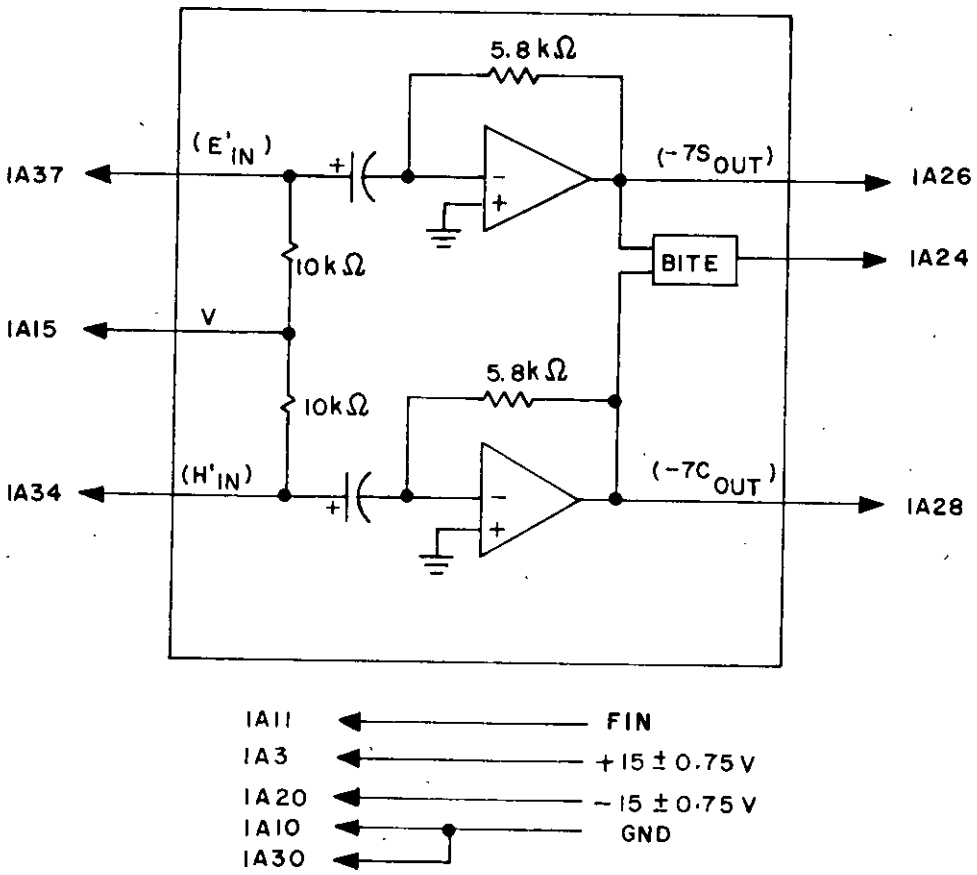
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be noninductive and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirement of MIL-M-28787.

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NOTE: The following pins are unused: 1A1, 1A2, 1A4, 1A5, 1A6, 1A7, 1A8, 1A9, 1A12, 1A13, 1A14, 1A16, 1A17, 1A18, 1A19, 1A21, 1A22, 1A23, 1A25, 1A27, 1A29, 1A31, 1A32, 1A33, 1A35, 1A36, 1A38, 1A39, and 1A40.

FIGURE 1. SES module function diagram, power amplifier.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A30	Isolation	II	S	10	∞	1.0	∞	10	∞	1.0	∞	MΩ
B1-B8	Hot socket	III	Q	← See table III and note 4 below →								
C	Stress	IV	Q	4.45	4.65	4.45	4.65	4.45	4.65	4.45	4.65	Vrms
D	Stress (V_S)	V	Q	7.8	8.2	7.8	8.2	7.8	8.2	7.8	8.2	Vrms
E1,E2	Output short circuit current (amplifier) check	VI	S	400	800	400	800	400	800	400	800	mVac
E3	Output short circuit current (BITE)	VI	S	0.3	1.0	0.3	1.0	0.3	1.0	0.3	1.0	mAdc
F1-F3	Accuracy (0°, 45°, 90°) 400 Hz	VII	X	← See table VII and note 4 below →								
F4,F5	Open circuit oscillation	VII	X	0	10	0	10	0	10	0	10	mV
F6,F7	Absolute gain (loaded) 400 Hz	VII	X	7.37	7.47	7.37	7.47	7.37	7.47	7.37	7.47	Vrms
G1,	Phase angle 400 Hz	VIII	X	179.5	180.5	179.5	180.5	179.5	180.5	179.5	180.5	De-grees
H1,H2	Quiescent power supply current	IX	S	0.5	50.0	0.5	50.0	0.5	50.0	0.5	50.0	mA
J1,J2	Output offset voltage	X	X	-4	4	-4	4	-4	4	-4	4	mV
K1-K4	BITE threshold check	XI	X	← See table XI and note 4 below →								
L1	Continuity, ground	XII	X	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	Ω
L2	Continuity, fin	XII	S	0	1	0	1	0	1	0	1	Ω
M1-M6	Line noise rejection 400 Hz	XIII	S	0	2	0	2	0	2	0	2	mV
N1,N2	Power supply transient amplitude	XIV	Q	0	300	0	300	0	300	0	300	mV
P1,P2	Power supply filter capacitance	XV	S	0.7	1.3	0.7	1.3	0.7	1.3	0.7	1.3	μF
Q	BITE output voltage, high level	XVI	X	2.4	5.5	2.4	5.5	2.4	5.5	2.4	5.5	Vdc
R1,R2	Distortion	XVII	S	0	1	0	1	0	1	0	1	%
S1,S2	Resistance	XVIII	X	9	11	9	11	9	11	9	11	kΩ

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ The initial and end-of-life limits for 25°C and operating temperature are the same.

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4.4.2 Order of testing. Test conditions A through S shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E, F, G where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through S in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions F through S shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test, electrical tests F through S shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/184-1 supersedes part number M28787/184. The two parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation	4 W (driving a 90 V L-L 5 kΩ synchro load with V 10 percent high)
Supply voltage	±15 Vdc ±5 percent maximum
Output offset voltage	4.0 mV maximum
Input (nominal)	1 Vrms +4.6 Vdc bias, 400 Hz ±10 percent The voltages are from a source resistance of 777Ω to ∞Ω.
Output voltage	7.4 Vrms, 400 Hz (nominal)
Output short circuit current (BITE)	0.6 A maximum (indicates a catastrophic failure only)

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Output voltage (BITE)	
High level ($I_{OH} \leq 200$ µA)	2.4 V minimum, 5.5 V maximum
Low level ($I_{OL} \leq 1.6$ mA)	0.0 minimum, 0.3 V maximum
Output drive (BITE)	
High level ($V_{OH} \geq 2.4$ V)	200 µA minimum
Low level ($V_{OL} \leq 0.4$ V)	3.2 mA maximum
Output load	40Ω minimum
Harmonic distortion	1 percent maximum

6.5 Module function. This module contains two independent power operational amplifiers and "BITE" circuitry for functional verification used in digital/synchro converter applications.

6.6 Input to output relationship. Each amplifier is configured in the inverting configuration. The BITE output indicates the presence or absence of amplifier outputs. The relationship for the BITE output is as follows:

High level	One or both present.
Low level	Both missing (failure).

The BITE only indicates a catastrophic failure. Both outputs have to be below 1.4 Vrms. The transition time high to low and low to high are not specified.

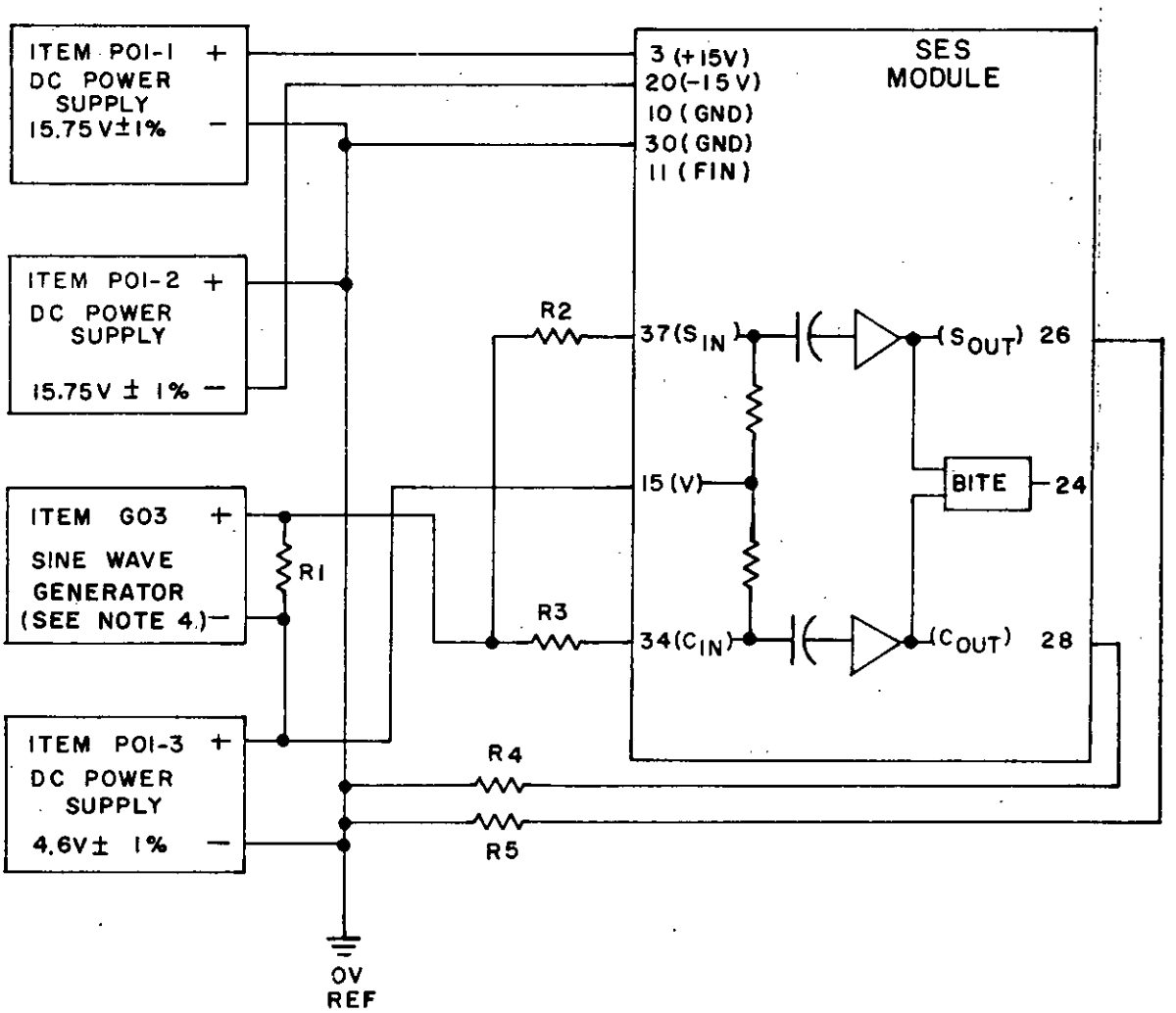
6.7 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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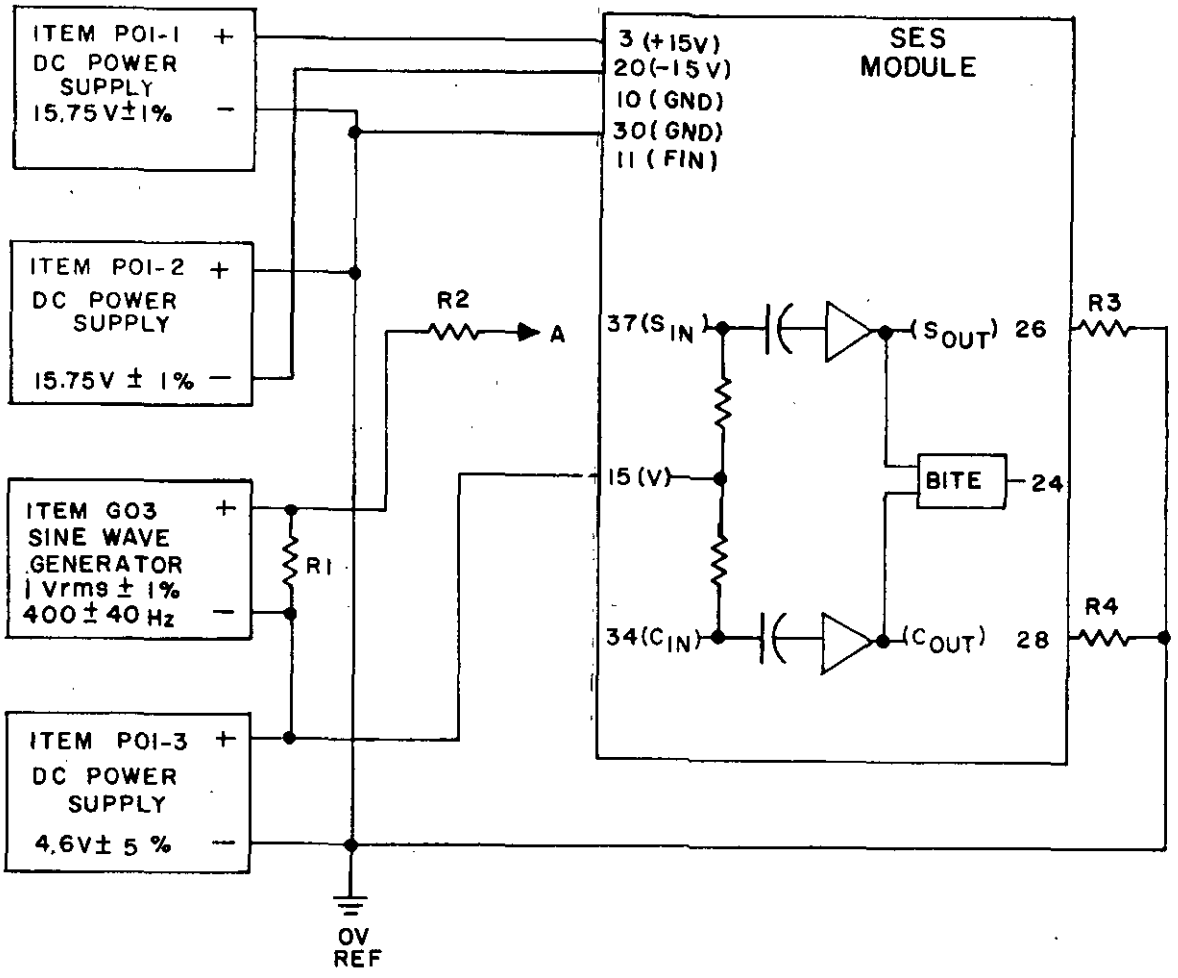
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NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2 = R3 = 5 kΩ ±5%, 1/4 W.
3. R4 = R5 = 50Ω ±5%, 2 W.
4. Adjust item G03 for an output of 7.6 Vrms ±10%, 400 Hz, at pin 26.

FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 777Ω ±1%, 1/8 W.
3. R3 = R4 = 40Ω ±1%, 2 W.
4. Module signal input.
   (a) For condition "a" connect point A to pin 34.
   (b) For condition "b" connect point A to pin 37.

FIGURE 3. Component temperature test circuit.

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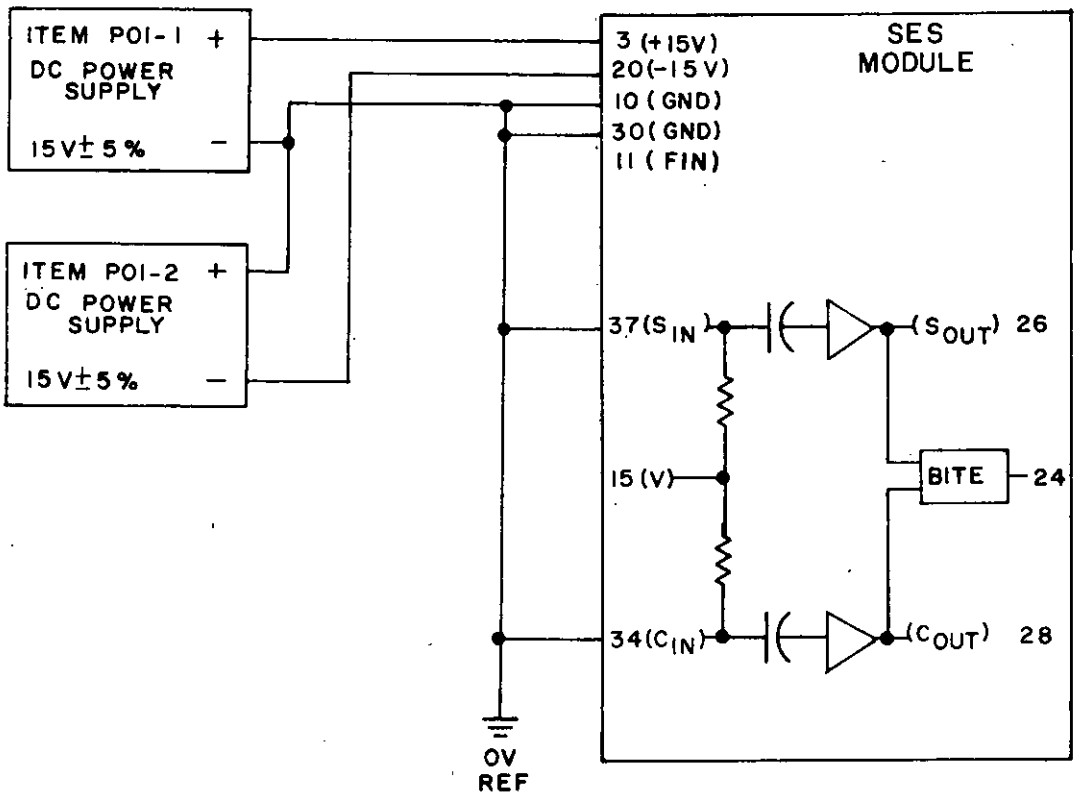


FIGURE 4. Vibration test circuit.

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TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation, fin pin to pins	C,1,2,4,5,6,7,8,9,12,13,14,16,17,18,19,21,22,23,25,27,29,31,32,33,35,36,38,39,40	11	10	∞	MΩ	5
A2	Isolation used pins to unused pins	1,2,4,5,6,7,8,9,12,13,14,16,17,18,19,21,22,23,25,27,29,31,32,33,35,36,38,39,40	C				
A3	Isolation, unused pins to unused pins	2,4,5,6,7,8,9,12,13,14,16,17,18,19,21,22,23,25,27,29,31,32,33,35,36,38,39,40	1				
A4	Isolation, unused pins to unused pins	4,5,6,7,8,9,12,13,14,16,17,18,19,21,22,23,25,27,29,31,32,33,35,36,38,39,40	2				
A5	Isolation, unused pins to unused pins	5,6,7,8,9,12,13,14,16,17,18,19,21,22,23,25,27,29,31,32,33,35,36,38,39,40	4				
A6	Isolation, unused pins to unused pins	6,7,8,9,12,13,14,16,17,18,19,21,22,23,25,27,29,31,32,33,35,36,38,39,40	5				
A7	Isolation, unused pins to unused pins	7,8,9,12,13,14,16,17,18,19,21,22,23,25,27,29,31,32,33,35,36,38,39,40	6				

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TABLE II. Isolation test - Continued.

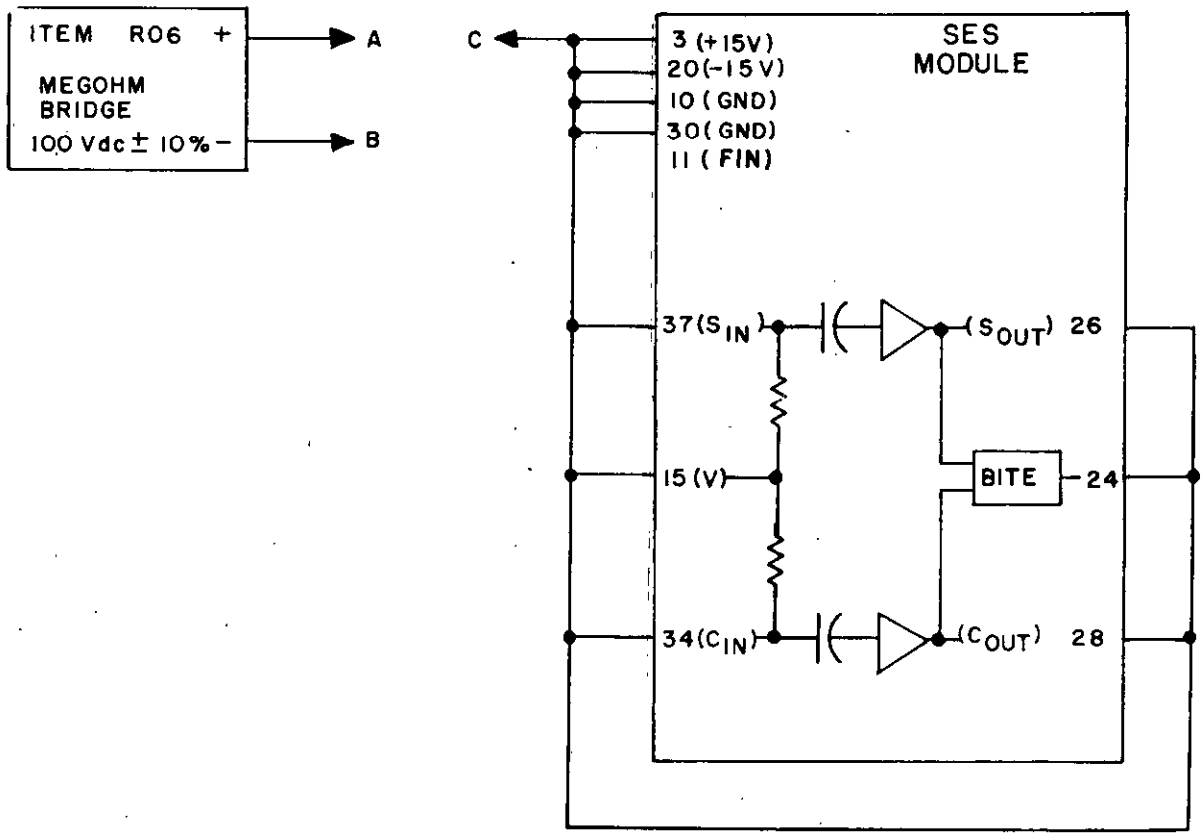
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A8	Isolation, unused pins to unused pins	8,9,12,13,14,16,17,18,19,21,22,23,25,27,29,31,32,33,35,36,38,39,40	7	10	∞	MΩ	5
A9	Isolation, unused pins to unused pins	9,12,13,14,16,17,18,19,21,22,23,25,27,29,31,32,33,35,36,38,39,40	8				
A10	Isolation, unused pins to unused pins	12,13,14,16,17,18,19,21,22,23,25,27,29,31,32,33,35,36,38,39,40	9				
A11	Isolation, unused pins to unused pins	13,14,16,17,18,19,21,22,23,25,27,29,31,32,33,35,36,38,39,40	12				
A12	Isolation, unused pins to unused pins	14,16,17,18,19,21,22,23,25,27,29,31,32,33,35,36,38,39,40	13				
A13	Isolation, unused pins to unused pins	16,17,18,19,21,22,23,25,27,29,31,32,33,35,36,38,39,40	14				
A14	Isolation, unused pins to unused pins	17,18,19,21,22,23,25,27,29,31,32,33,35,36,38,39,40	16				
A15	Isolation, unused pins to unused pins	18,19,21,22,23,25,27,29,31,32,33,35,36,38,39,40	17				
A16	Isolation, unused pins to unused pins	19,21,22,23,25,27,29,31,32,33,35,36,38,39,40	18				
A17	Isolation, unused pins to unused pins	21,22,23,25,27,29,31,32,33,35,36,38,39,40	19				

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TABLE II. Isolation test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A18	Isolation, unused pins to unused pins	22,23,25,27,29,31,32,33, 35,36,38,39,40	21	10	∞	MΩ	5
A19	Isolation, unused pins to unused pins	23,25,27,29,31,32,33, 35,36,38,39,40	22				
A20	Isolation, unused pins to unused pins	25,27,29,31,32,33,35, 36,38,39,40	23				
A21	Isolation, unused pins to unused pins	27,29,31,32,33,35,36, 38,39,40	25				
A22	Isolation, unused pins to unused pins	29,31,32,33,35,36,38, 39,40	27				
A23	Isolation, unused pins to unused pins	31,32,33,35,36,38,39,40	29				
A24	Isolation, unused pins to unused pins	32,33,35,36,38,39,40	31				
A25	Isolation, unused pins to unused pins	33,35,36,38,39,40	32				
A26	Isolation, unused pins to unused pins	35,36,38,39,40	33				
A27	Isolation, unused pins to unused pins	36,38,39,40	35				
A28	Isolation, unused pins to unused pins	38,39,40	36				
A29	Isolation, unused pins to unused pins	39,40	38				
A30	Isolation, unused pins to unused pins	40	39				

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

FIGURE 5. Isolation test circuit for table II.

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TABLE III. Hot socket test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
B1	Hot socket	26	4.43	4.63	Vrms	6
B2		28	4.43	4.63	Vrms	
B3		Open	2.5	5.5	Vdc	
B4		Open	0.0	0.4	Vdc	
B5		26	4.43	4.63	Vrms	
B6		28	4.43	4.63	Vrms	
B7		Open	0.0	0.4	Vdc	
B8		Open	2.5	5.5	Vdc	

TABLE IV. Stress test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C	Stress	26	4.45	4.65	Vrms	8

TABLE V. Stress (V_s) test.

Test condition	Parameter	Connect A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
D	Stress (V_s)	Open	37	34	28	26	28	Open	7.8	8.2	Vrms	9

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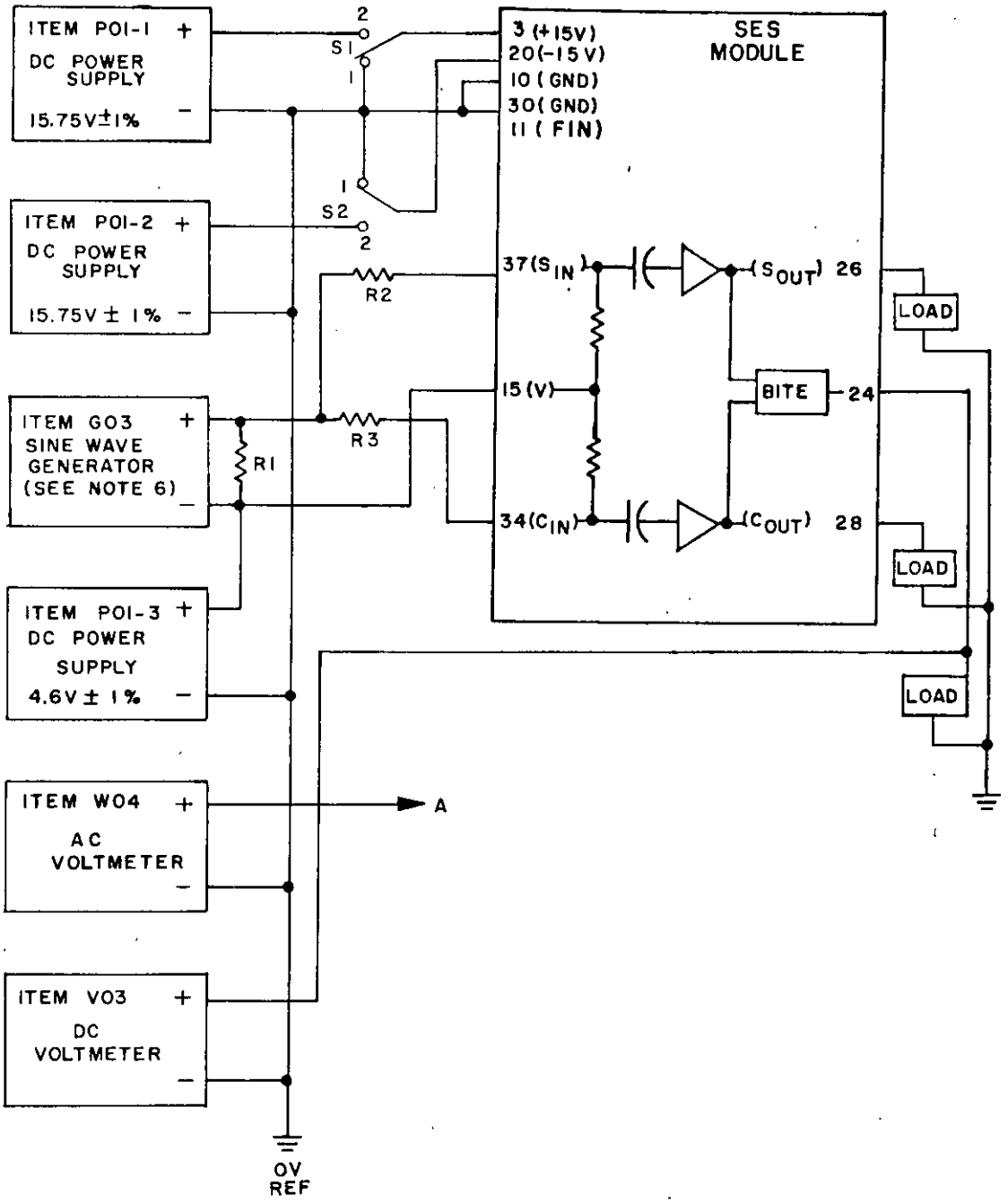
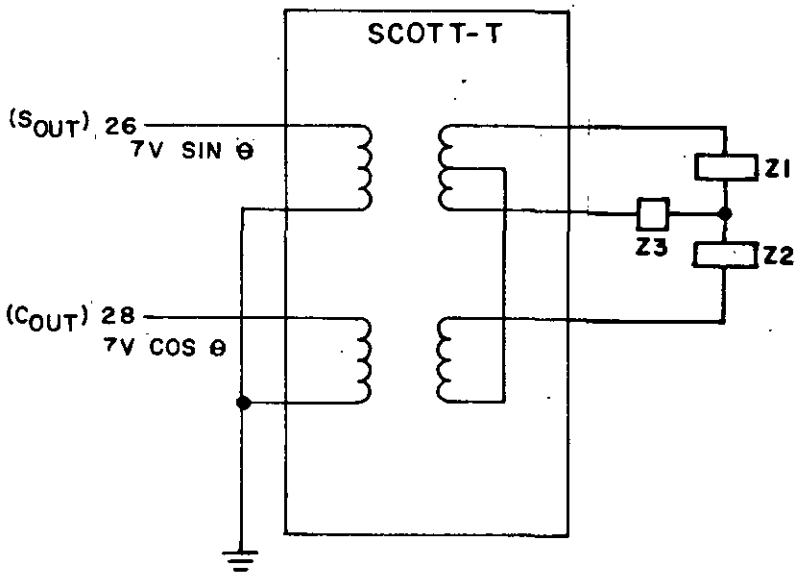


FIGURE 6. Hot socket test circuit for table III.

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NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2 = R3 = 1,277Ω ±1%, 1/8 W matched to 0.005%.
3. S1 = S2 = SPDT switch.
4. Load BITE (pin 24) as specified on figure 7.
5. Load pin 26 and pin 28 as follows:



Z1-Z3 = 3-1/3 kΩ ±1%
with $Q = \frac{\omega L}{R} = 5 \pm 1\%$
where f = 400 Hz,
$\omega = 2\pi f$,
Q matched to 1%.
Output Scott-T: 90 V line-to-line synchro signal out with 7 V resolver signal in.

6. Adjust item G03 for 400 ±4 Hz with an amplitude as specified below:

   Amplitude = 0.3 Vrms ±1% for test conditions B4 and B7.
   Amplitude = 1 ±0.010 Vrms for test conditions B1, B2, B3, B5, B6, and B8.

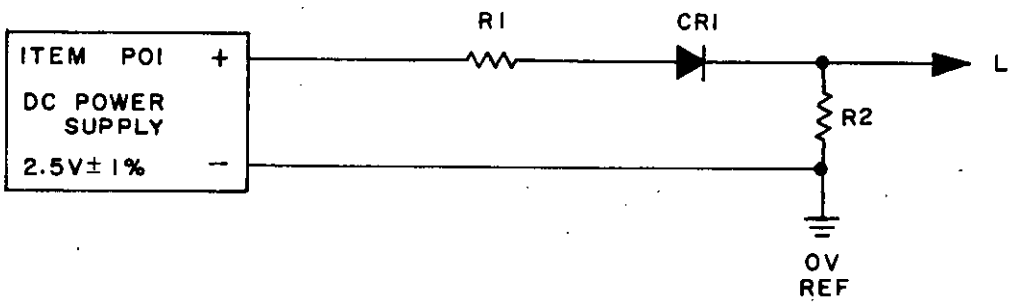
7. Switching sequence.

   (a) Initially place switch S1 and switch S2 in position 1. Operate switch S1 to position 2 then place switch S2 to position 2. Perform test conditions B1, B2, B3, and B4.

   (b) Reset switch S1 and switch S2 to position 1. Wait 10 seconds then operate switch S2 to position 2, followed by switch S1 to position 2. Perform test conditions B5, B6, B7, and B8.

FIGURE 6. Hot socket test circuit for table III - Continued.

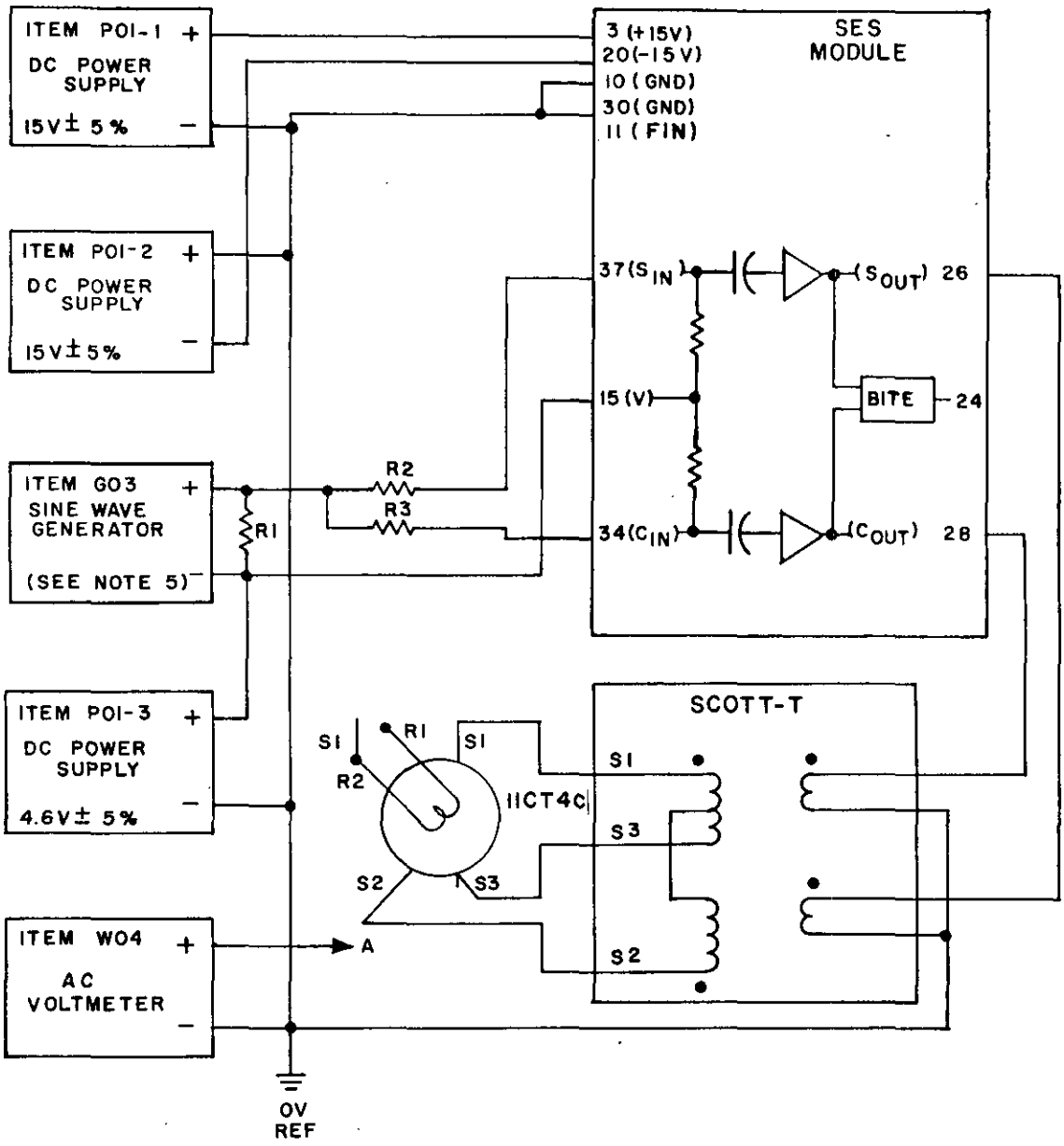
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NOTES:
1. R1 = Select value for a current flow of 1.6 mA ±5% when point L is held at 0.4 V.
2. R2 = 12 kΩ ±1%, 1/4 W (current flow of 200 μA when point L is held at 2.4 V).
3. CR1 = 1N914, or equivalent.

FIGURE 7. Load for BITE output.

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NOTES:
1. R1 = 50Ω ±5%, 2 W.
2. R2 = R3 = 1,277Ω ±1%, 1/8 W matched to 0.005%,
3. S = SPST switch.
4. Output Scott-T, 7 V resolver signal in, 90 V L-L synchro signal out.
5. Adjust item G03 for 400 ±4 Hz with an amplitude of 1.0 Vrms ±1%.
6. Open switch S1, position shaft of synchro 11CT4c for maximum voltage reading across synchro terminals R1-R2.
7. After adjustment open switch S1 for 10 to 30 seconds then close. Read voltage on item W04.

FIGURE 8. Stress test circuit for table IV.

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TABLE VI. Output short circuit current-test.

Test condition	Parameter	Connect							Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
E1	Output short circuit current (amplifier) check	Open	37	34	26,H	Open	Open	Open	400	800	mVac	9
E2	Output short circuit current (amplifier) check	Open	37	34	28,H	Open	Open	Open	400	800	mVac	9
E3	Output short circuit current (BITE)	Open	37	34	Open	Open	Open	24	0.3	1.0	mAdc	9

*TABLE VII. Accuracy, open circuit oscillation, and absolute gain test.

Test condition	Parameter	Connect							Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
F1	Accuracy (90°) 400 Hz	37	Open	Open	Open	26	28	Open	89.9850	90.0150	Degrees	9
F2	Accuracy (45°) 400 Hz	Open	37	34	Open	26	28	Open	44.98	45.02	Degrees	
F3	Accuracy (0°) 400 Hz	34	Open	Open	Open	26	28	Open	359.9850	0.0150	Degrees	
F4	Open circuit oscillation	Open	Open	Open	28	Open	Open	Open	0	10	mV	
F5	Open circuit oscillation	Open	Open	Open	26	Open	Open	Open	0	10	mV	
F6	Absolute gain (loaded) 400 Hz	37	Open	Open	26	26	28	Open	7.37	7.47	Vrms	
F7	Absolute gain (loaded) 400 Hz	34	Open	Open	28	26	28	Open	7.37	7.47	Vrms	

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*TABLE VIII. Phase angle test.

Test condition	Parameter	Connect							Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
G1	Phase angle (400 Hz)	37	Open	Open	Open	26	28	Open	179.5	180.5	Degrees	9

TABLE IX. Quiescent power supply current test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
H1	Quiescent power supply current (-15.75 V)	0.5	50.0	mA	10
H2	Quiescent power supply current (+15.75 V)	0.5	50.0	mA	10

TABLE X. Output offset voltage test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
J1	Output offset voltage	26	-4	4	mV	11
J2	Output offset voltage	28	-4	4	mV	11

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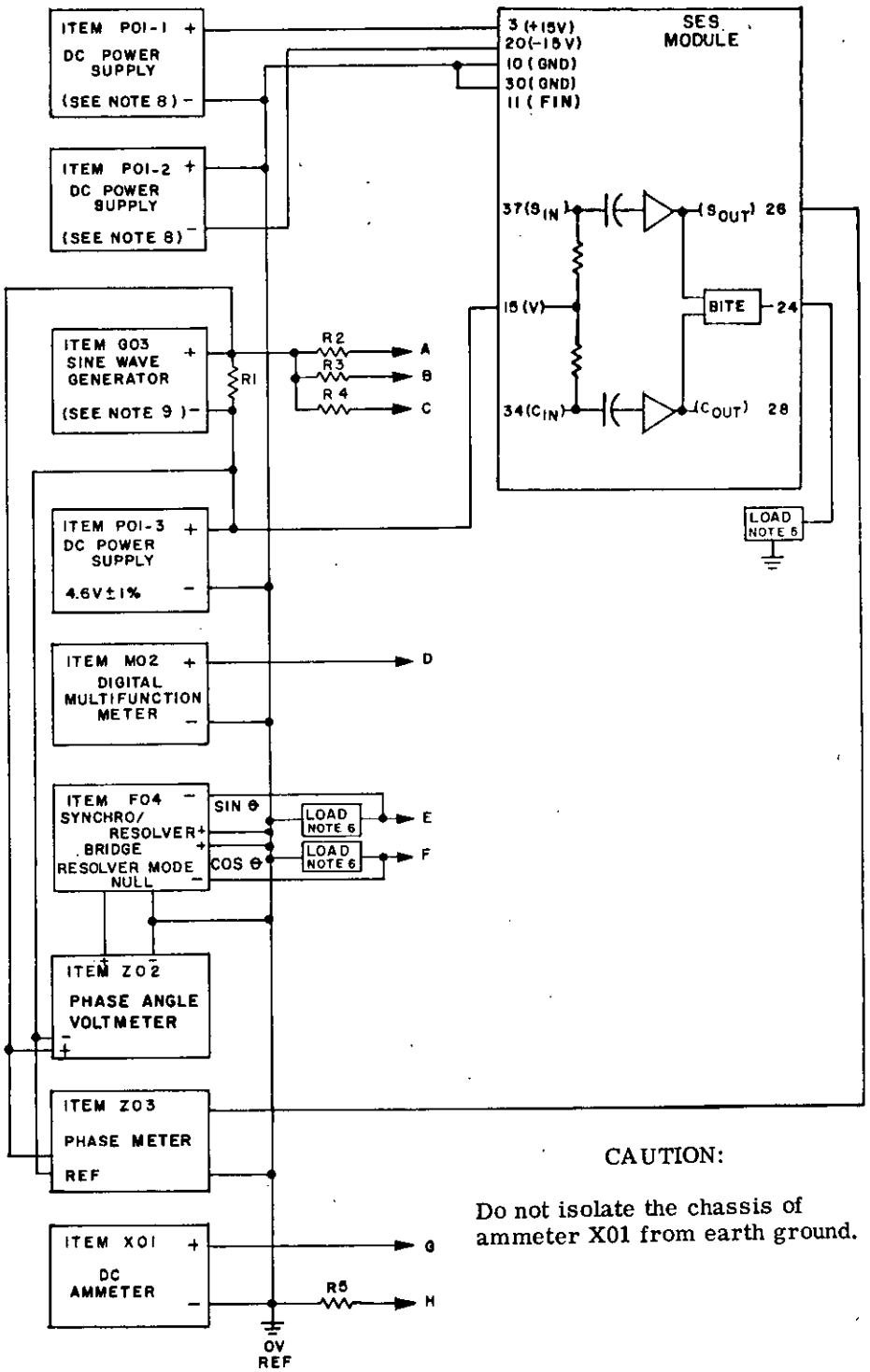
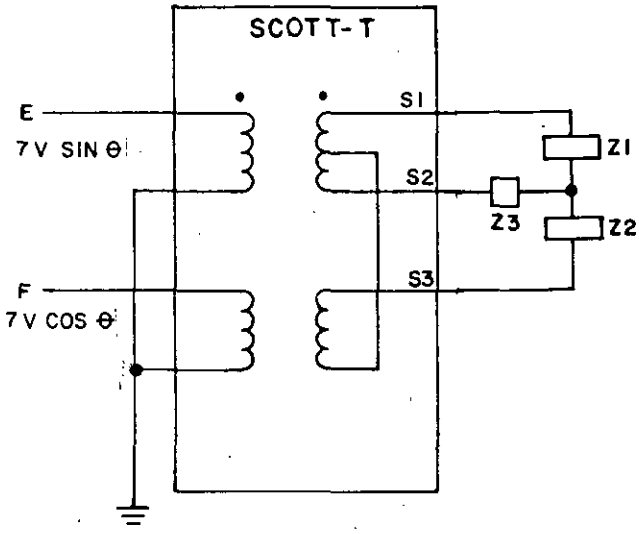


*FIGURE 9. Stress (V_S), output short circuit current, accuracy, open circuit oscillation, absolute gain, and phase angle test circuit for tables V, VI, VII, and VIII.

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NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2 = 777Ω ±0.1%, 1/8 W.
3. R3 = R4 = 1,277Ω ±1%, 1/8 W matched to 0.005%.
4. R5 = 1Ω ±1%, 1/2 W.
5. Load BITE (pin 24) as specified on figure 7.
6. Load (400 Hz):



Z1-Z3 = 3 1/3 kΩ ±1%
with $Q = \frac{\omega L}{R} = 5$ ±1%
where f = 400 Hz, ω = 2 πf,
Q matched to 1%.
Output Scott-T: 90 V line-to-line synchro signal out with 7 V resolver signal in.

7. Do test conditions D, E, F, and G without disconnecting items P01-1 and P01-2 and never decreasing the voltage below 14.25 V.
8. Adjust items P01-1 and P01-2 as follows:

Test condition	P01-1	P01-2
D	18 V ±1%	18 V ±1%
E1-E3	15.75 V ±1%	15.75 V ±1%
F1-F7	15 V ±1%	15 V ±1%
G1	15 V ±1%	15 V ±1%

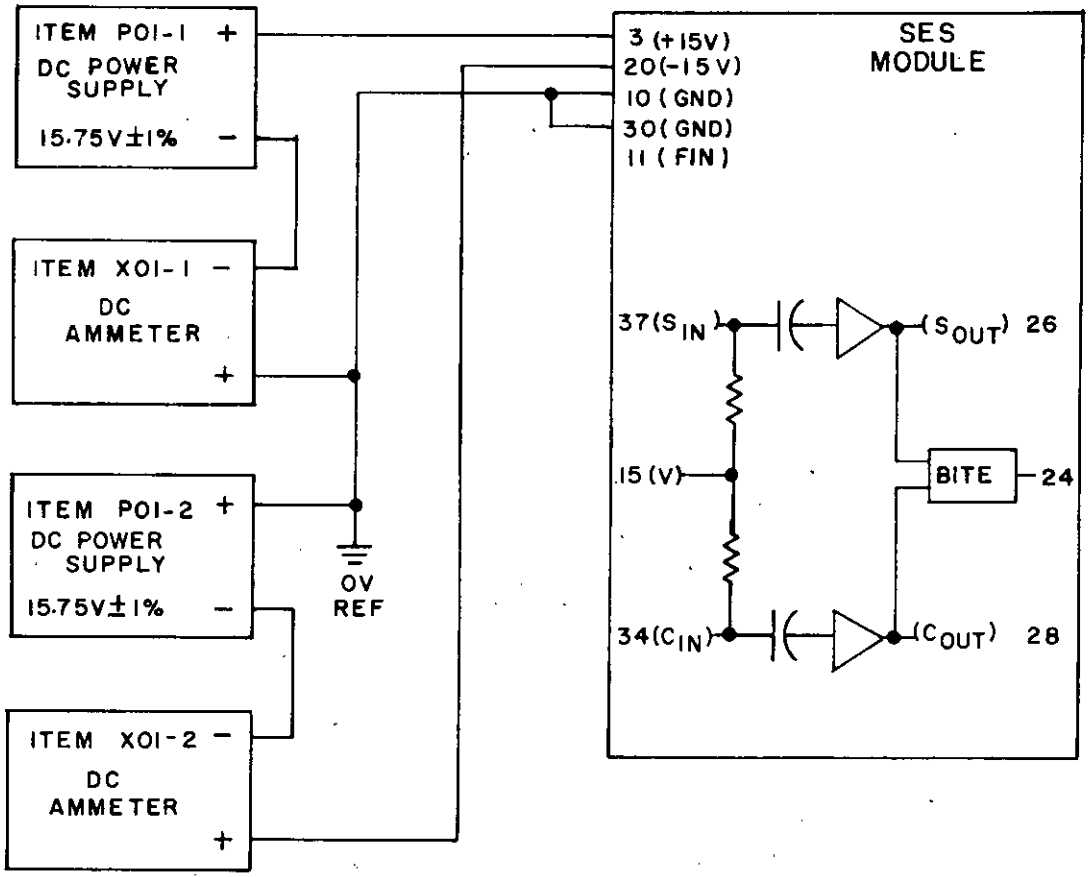
9. Adjust item G03 as follows:

Test condition	
D	For an output on pin 26 of 8.0 ±0.1 Vrms, 400 ±4 Hz.
E1-E3	For an output on pin 26 of 8.0 ±0.1 Vrms, 400 ±4 Hz.
F1-F5	1.0 ±0.1 Vrms, 400 ±4 Hz at item G03 output.
F6, F7	1.000 ±0.001 Vrms, 400 ±4 Hz at item G03 output.
G1	1.000 ±0.001 Vrms, 400 ±4 Hz at item G03 output.

10. When performing output short circuit current test, only one output shall be shorted at a time and the measurement taken after 5 seconds but before 10 seconds.
11. For test condition E3 the load on pin 24 should be removed.
12. Readings for test conditions E1 and E2 can be read directly from item M02 since $I = E/R$ and R = 1Ω.
13. Take reading for test condition G1 following test condition F6.

FIGURE 9. Stress (V_S), output short circuit current, accuracy, open circuit oscillation, absolute gain, and phase angle test circuit for tables V, VI, VII, and VIII - Continued.

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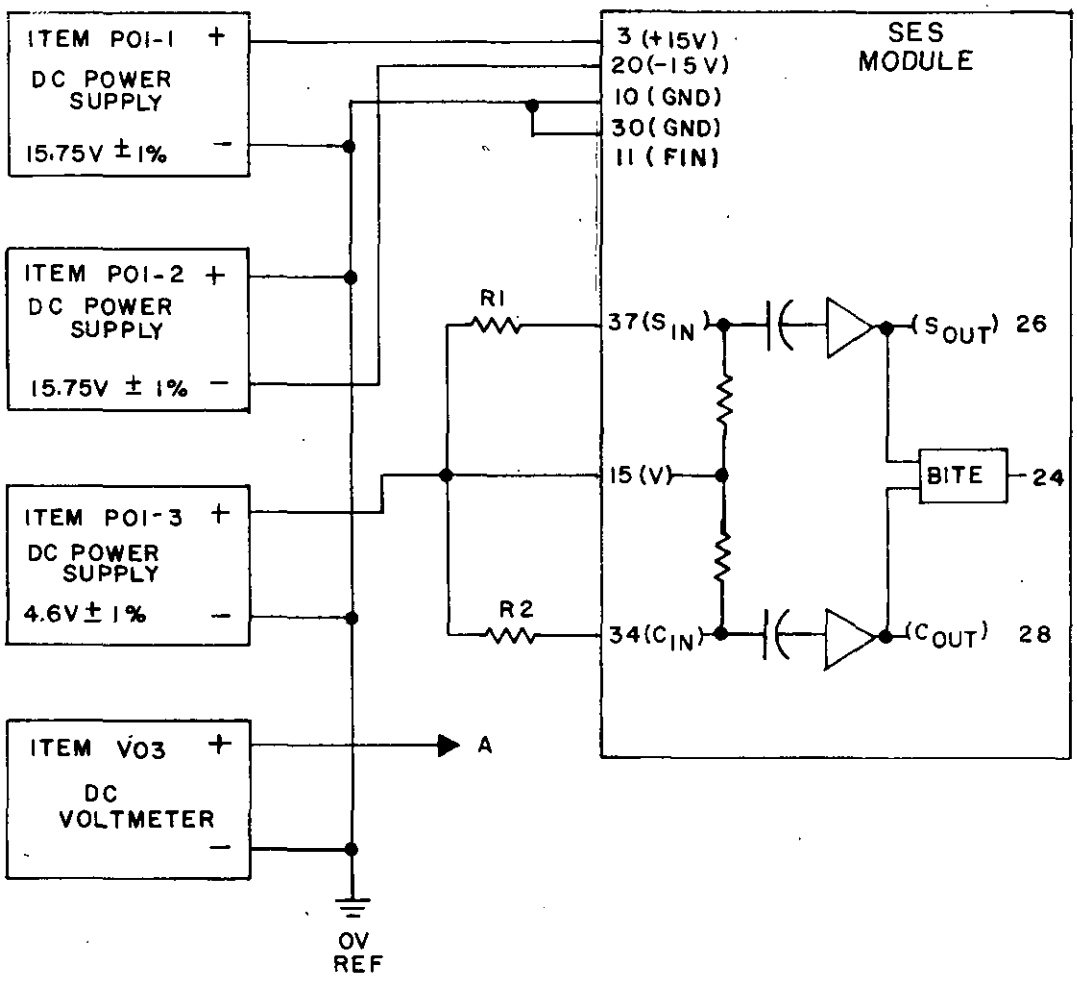


CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 10. Quiescent power supply current test circuit for table IX.

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NOTE: R1 = R2 = 50Ω ±5%, 1/8 W.

FIGURE 11. Output offset voltage test circuit for table X.

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TABLE XI. BITE threshold check.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	BITE threshold, low level, check	37	34	0.0	0.4	V	12
K2	BITE threshold, high level, check	37	34	2.4	5.0		
K3	BITE threshold, low level, check	37	Open	0.0	0.4		
K4	BITE threshold, low level, check	Open	34	0.0	0.4		

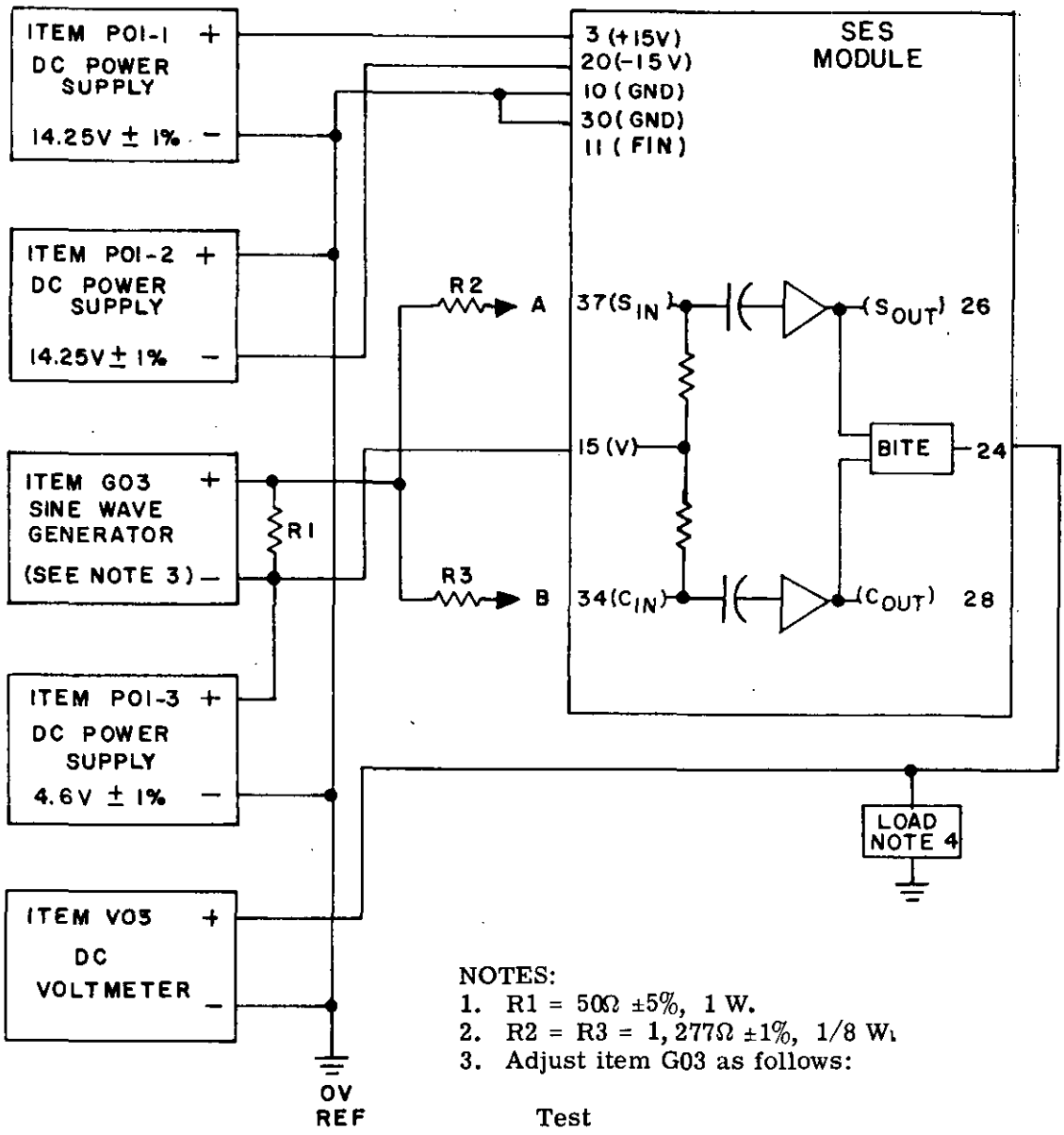
TABLE XII. Continuity test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L1	Continuity, ground	10	30	0.0	0.1	Ω	13
L2	Continuity, pin	11	Fin	0	1	Ω	13

TABLE XIII. Line noise rejection (400 Hz) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Line noise rejection, +15 V	C	26	0	2	mV	14
M2	Line noise rejection, +15 V	C	28				
M3	Line noise rejection, -15 V	D	26				
M4	Line noise rejection, -15 V	D	28				
M5	Calculation (M1 + M3)	---	---				
M6	Calculation (M2 + M4)	---	---				

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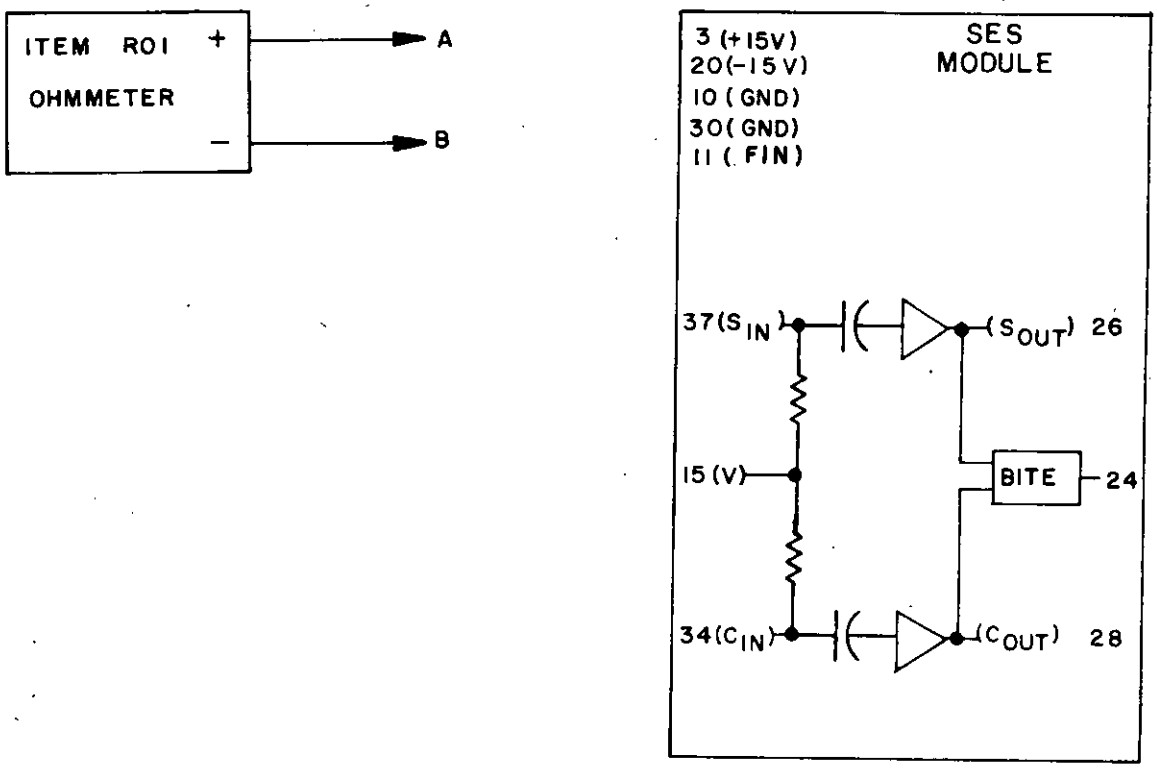
NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. R2 = R3 = 1,277Ω ±1%, 1/8 W.
3. Adjust item G03 as follows:

	Test condition
K1	0.3 Vrms ±1%, 400 ±4 Hz.
K2	0.7 Vrms ±1%, 400 ±4 Hz.
K3, K4	0.5 Vrms ±1%, 400 ±4 Hz.

4. Load BITE (pin 24) as specified on figure 7.
5. This test verifies BITE output drive.

FIGURE 12. BITE threshold check test circuit for table XI.

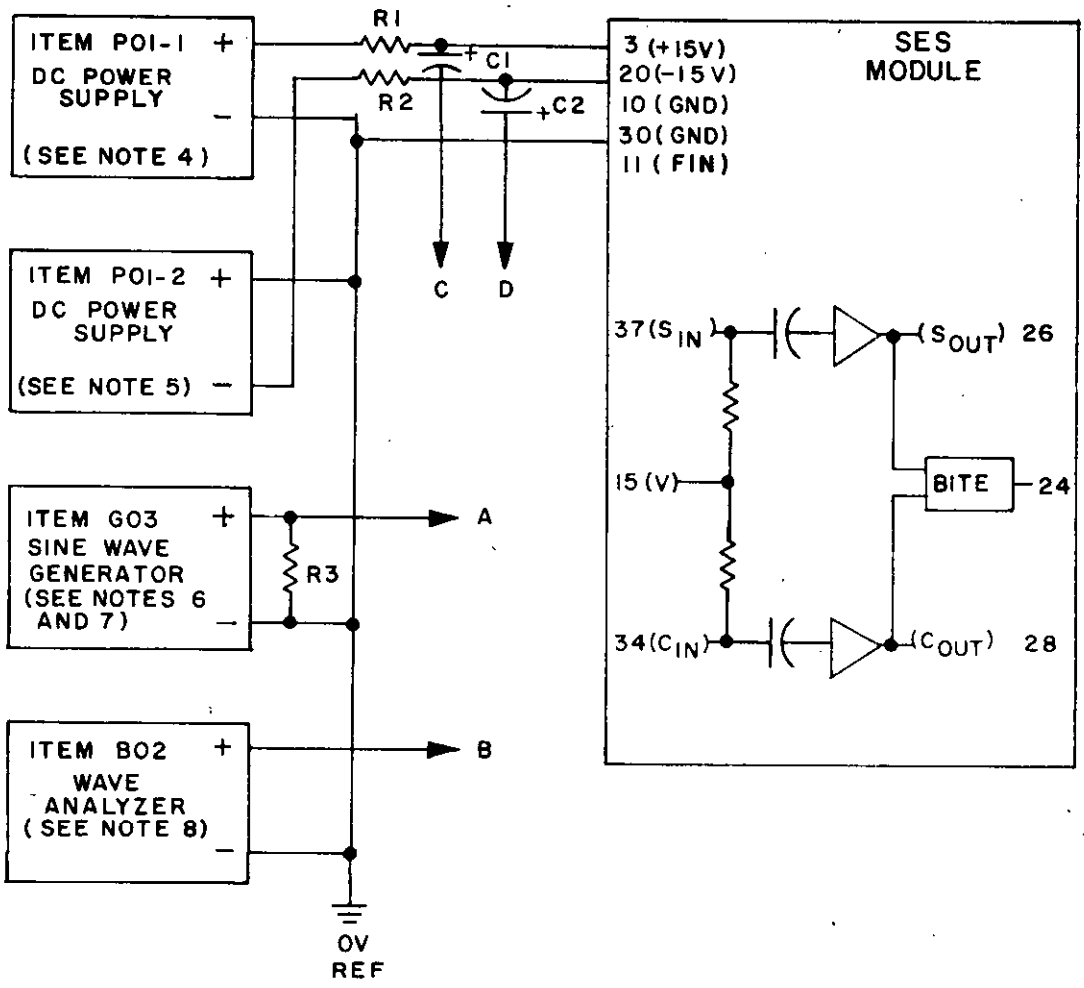
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NOTE: On test condition L2 (when measuring to fin), penetrate the anodized finish of the module fin with a sharply pointed probe connected to the negative terminal.

FIGURE 13. Continuity test circuit for table XII.

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NOTES:

1. R1 = R2 = 200Ω ±5%, 5 W.
2. R3 = 600Ω ±5%, 1 W.
3. C1 = C2 = 100 μF ±10%, 20 WVdc, electrolytic.
4. Adjust item P01-1 for +15 Vdc ±1% on pin 3.
5. Adjust item P01-2 for -15 Vdc ±1% on pin 20.
6. For test conditions M1 and M2 adjust item G03 for 400 ±4 Hz and 1 Vrms ±1% on pin 3.
7. For test conditions M3 and M4 adjust item G03 for 400 ±4 Hz and 1 Vrms ±1% on pin 20.
8. For test conditions M1-M4 adjust item B02 for 400 Hz voltage readings.
9. For test condition M5 the sum of the voltage readings from test conditions M1 and M3 shall be equal to or less than 2 mV (M1 + M3 $\leq$ 2 mV).
10. For test condition M6 the sum of the voltage readings from test conditions M2 and M4 shall be equal to or less than 2 mV (M2 + M4 $\leq$ 2 mV).

FIGURE 14. Line noise rejection (400 Hz) test circuit for table XIII.

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TABLE XIV. Power supply transient amplitude test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		B to pin	Min	Max		
N1	Power supply transient amplitude (+15 V)	3	0	300	mV	15,16
N2	Power supply transient amplitude (-15 V)	20	0	300	mV	15,16

TABLE XV. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
P1	Power supply filter capacitance	3	10	0.7	1.3	µF	17
P2	Power supply filter capacitance	10	20	0.7	1.3	µF	17

TABLE XVI. BITE output voltage (high-level) test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
Q	BITE output voltage, high level	2.4	5.5	Vdc	18

TABLE XVII. Distortion test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
R1	Distortion	37	26	0	1	%	19
R2	Distortion	34	28	0	1	%	19

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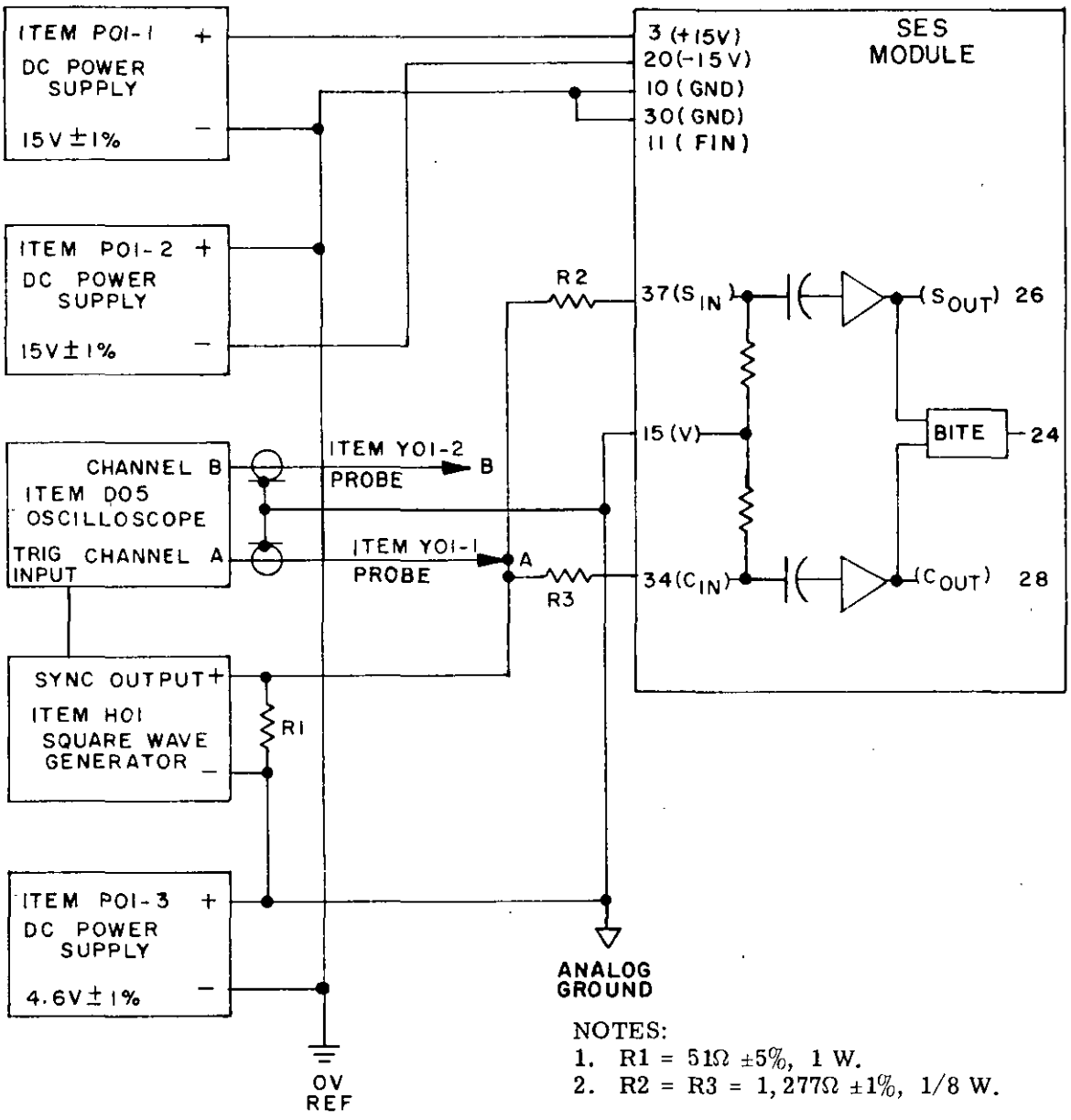
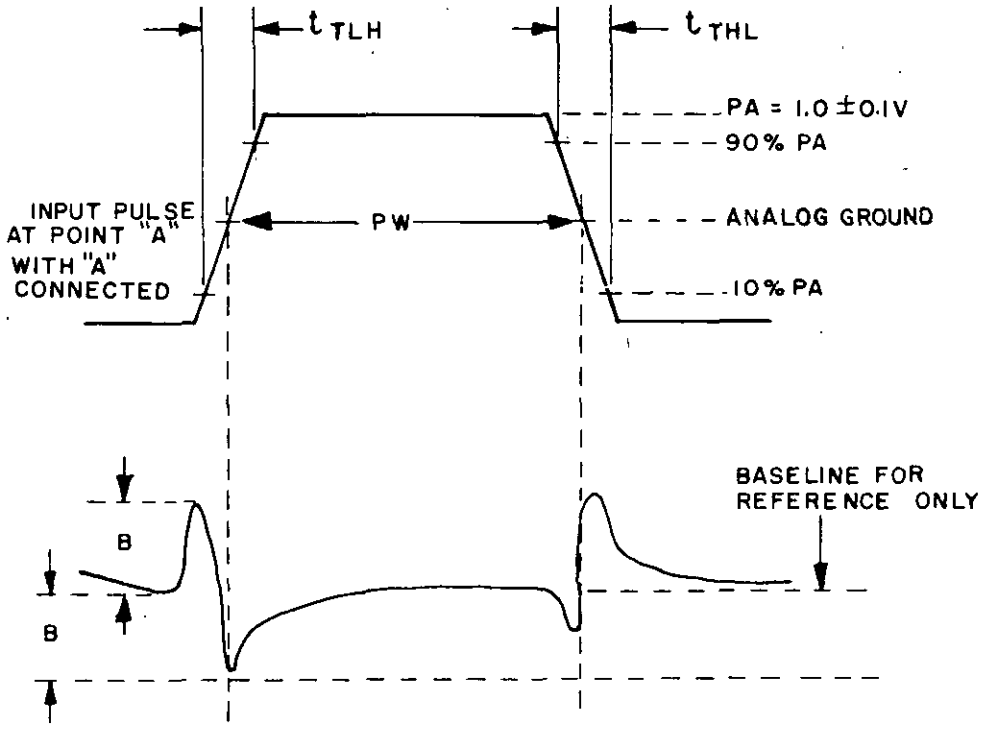


FIGURE 15. Power supply transient amplitude test circuit for table XIV.

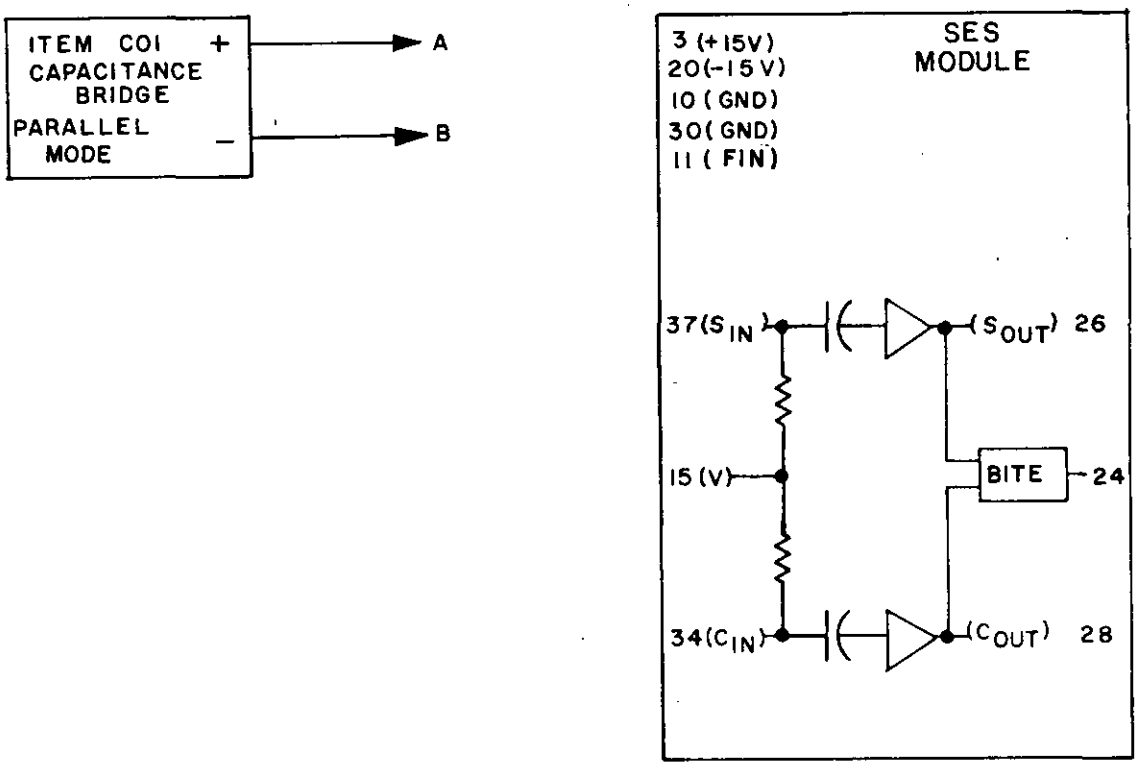
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NOTES:
1. $t_{TLH} = t_{THL} \leq 1$ µs.
2. PW = 1.25 ms ±10%.
3. PRP = 2.5 ms ±10%.
4. For test condition N1, B is measured from the baseline to the most negative excursion of the waveform.
5. For test condition N2, B is measured from the baseline to the most positive excursion of the waveform.
6. All voltage levels are referenced to ground.

FIGURE 16. Power supply transient amplitude test circuit waveforms for table XIV.

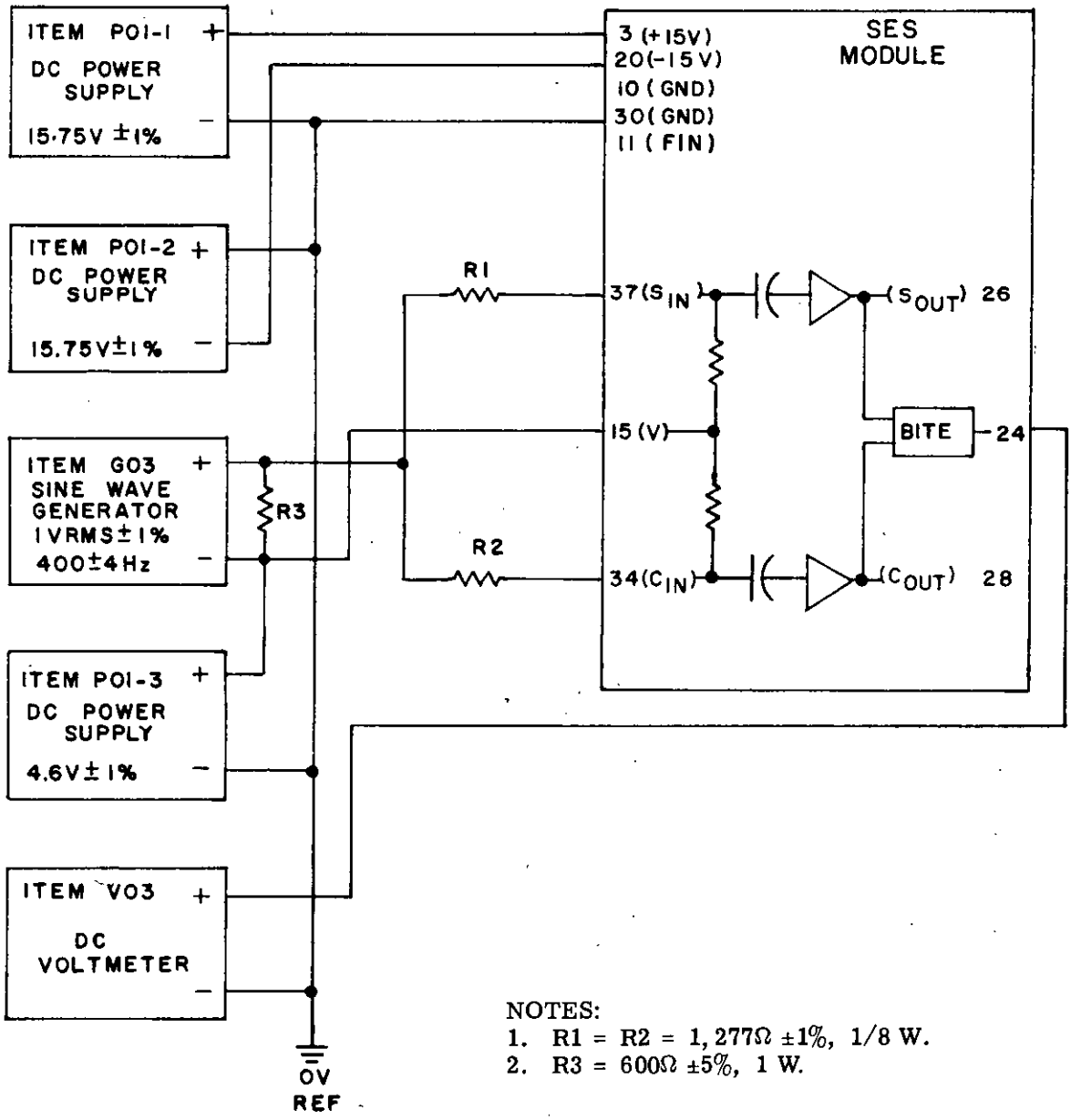
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NOTE: Capacitance bridge (C01) setup:
Frequency = 120 Hz.
Vrms = 0.3 V maximum.

FIGURE 17. Power supply filter capacitance test circuit for table XV.

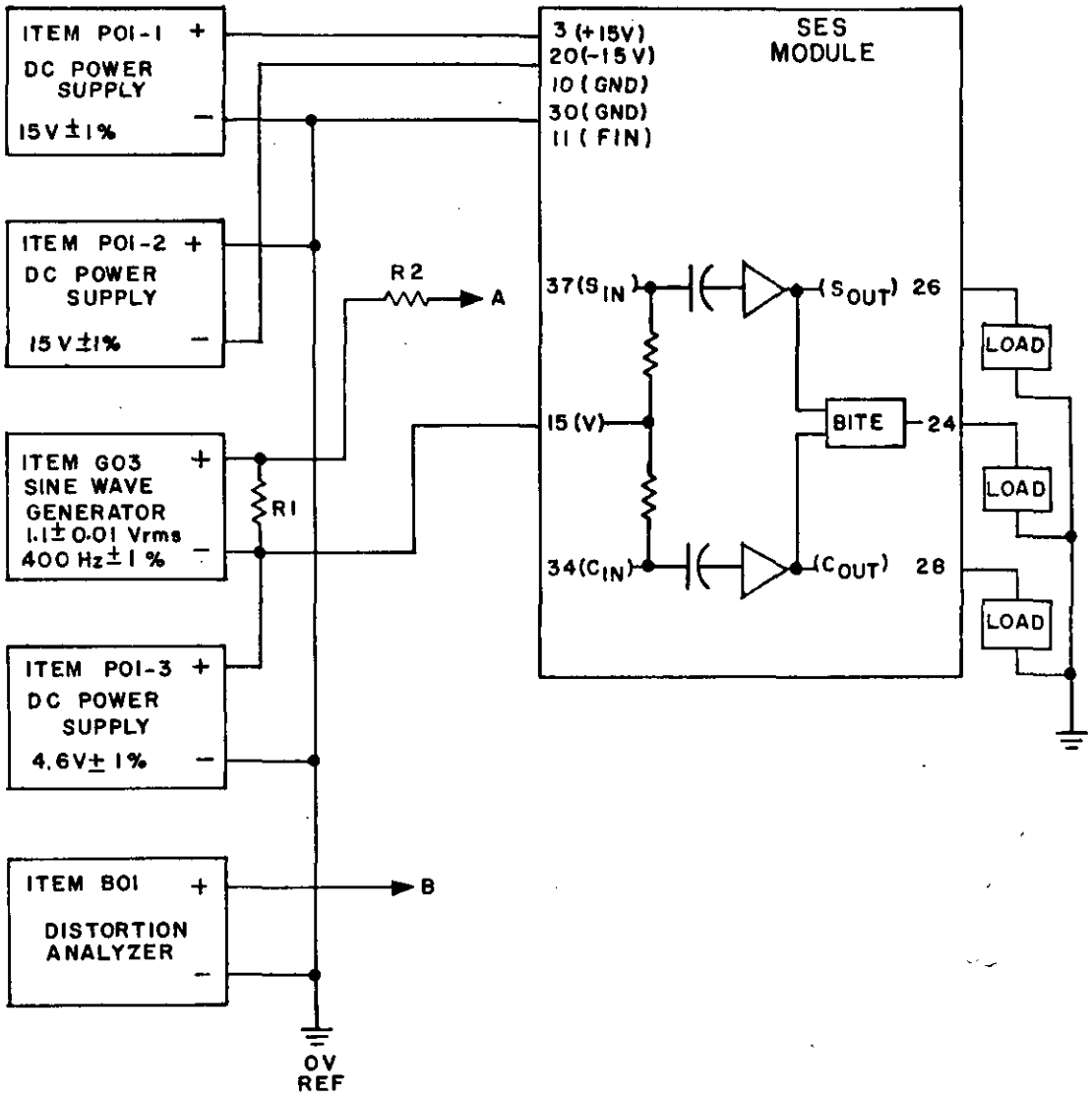
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NOTES:
1. R1 = R2 = 1,277Ω ±1%, 1/8 W.
2. R3 = 600Ω ±5%, 1 W.

FIGURE 18. BITE output voltage (high-level) test circuit for table XVI.

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NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2 = 777Ω ±0.1%, 1/8 W.
3. Load BITE (pin 24) as specified on figure 7.
4. Load pins 26 and 28 as specified on figure 9, note 6.

FIGURE 19. Distortion test circuit for table XVII.

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TABLE XVIII. Resistance test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
S1	Resistance	37	9	11	kΩ	20
S2	Resistance	34	9	11	kΩ	20

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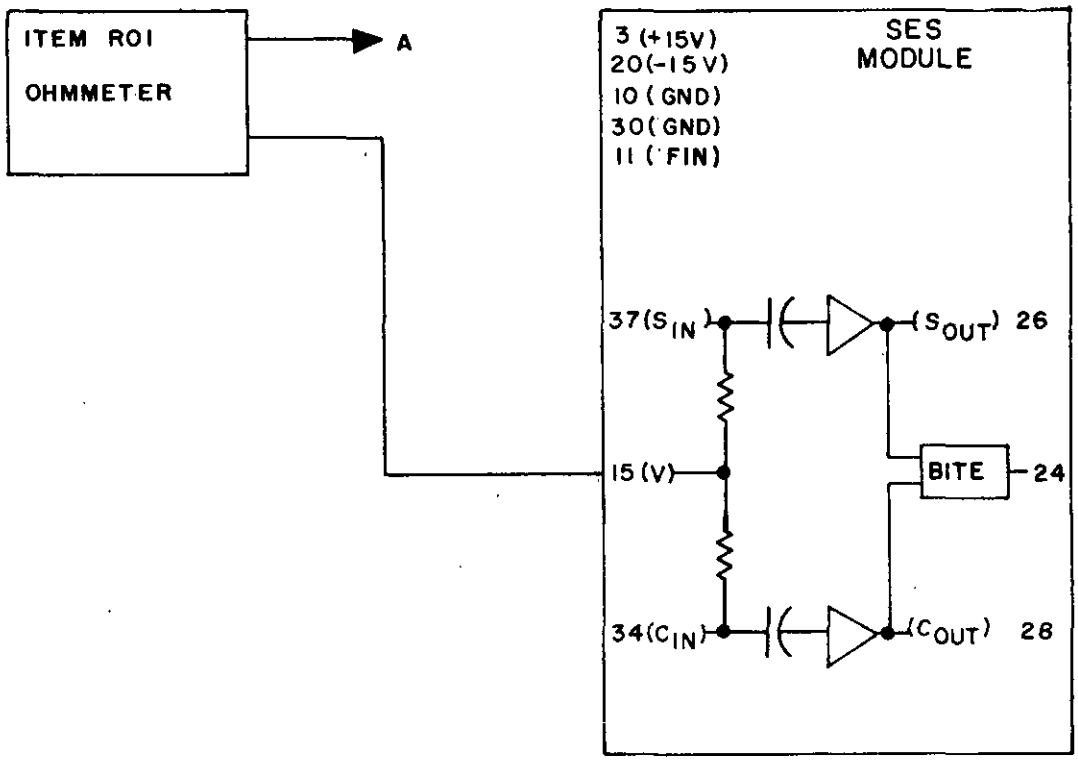


FIGURE 20. Resistance test circuit for table XVIII.

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